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Project Name: L01, Riyadh, Kingdom of Saudi Arabia

METHOD STATEMENT

FOR

CONCRETE WORK

MBLC				
Submitted By	Reviewed By	Verified By	Date	30-05-2020
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1. Scope

This method statement shall apply to all personnel involved in and who are responsible for the execution of concreting works.

2. Purpose

1.1. The purpose of this method statement is to describe the Construction methods, Material Submittals, Design Drawings, Equipment's, Tools, Hazard Analysis and the ITP for the concrete work as a part of the L01 Project.

1.2. This method statement will allow the construction entities (client and its subcontractor) to carry out the work specified under SCOPE OF WORK in a coordinated and organized manner, following a uniform approach by all parties involved. The information contained in this MES will be the governing guideline for all construction activities for the field personnel and everyone involved in the implementation of this phase of construction of this part of the Project.

1.3. It will also provide valuable information to the Engineer, who will be in position to follow and monitor efficiently the work activities in progress. Plan the forthcoming work activities.

3. Responsibility

An effective team shall be deployed to include both technical and expertise relevant to the facility at hand. It is imperative that the team is technically proficient with the installation techniques and tools necessary to execute the works.

The installation team must be structured, with defined roles and responsibilities as described below:

Position	Role	Responsibilities
Construction Manager	To ensure the schedule of works are on time.	He is overall responsible for ensuring that the formwork installation and all safety issues is carried out in accordance with this Method Statement. He is responsible to ensure that adequate manpower, the required material and the specified equipment are made available to the commencement of the works. He shall also ensure that proper access and

Position	Role	Responsibilities
		adequate lighting is provided in addition of all safety issues in site during the work. Prepare site work schedule as per site duty of work, based on project schedule. Instruct the Site Foreman for any exploratory excavation at site.
QC Engineer	Ensure the conformance of works done installer Identify defects and damages during work progress Detect possible quality problem	Prepare Check list. Prepare the WIR. He is responsible for ensuring that inspection is carried as required
HSE Engineer	Vigilant on-site monitoring The Safety Team maintains awareness of active and developing situations. The Safety Team ensures the Site Safety and Health Plan is prepared and implemented. The Safety Team ensures there are safety messages in each Incident Action Plan	The Safety Team is responsible for monitoring and assessing hazardous and unsafe situations and developing measures to assure personnel safety. The Safety Team will correct unsafe acts or conditions through the regular line of authority, although the Safety Team may exercise emergency authority to prevent or stop unsafe acts when immediate action is required. He is responsible to check that the appropriate measures are taken into account to ensure the work is undertaken in a safe manner
Site Engineer	To provide all necessary information and distribute responsibilities to his construction team. To co-ordinate with the safety officer and to ensure that the works are carried out in safe practicing methods.	The site engineer will ensure that the site preparation activities are conducted according to the approval method statement. It is the site supervisor s responsibility to organize the work in a safe manner to ensure that safety and protective equipment are being properly utilized. He must also ensure that the safety requirements are implemented and the approved method statement procedure is followed and the ITP is applied

Position	Role	Responsibilities
Foreman	To ensure in consultation with the site engineer that the manpower involved in the works are moving as agreed and planned for the work.	The foreman will carry out his duties by maintaining continuous coordination with the site engineer on daily basis, and ensure proper distribution of the workforce in the required and planned locations. Ensure that his technicians are aware of the job requirements and they have enough information essential tools and required materials to carry out their duties properly.
Technician	To ensure in consultation with the Foreman that the manpower involved in the works are moving as agreed and planned for the work.	The Technician will carry out his duties by maintaining continuous coordination with the foreman on daily basis, and ensure proper distribution of the workforce in the required and planned locations. Ensure that his assistant foreman/charge hands are aware of the job requirements and they have enough information essential tools and required materials to carry out their duties properly.

4. Procedure

4.1 MATERIALS

- Concrete shall be designed and prepared to meet the quality requirements. The mix design shall be submitted to the Designer for approval at least fifteen (15) days prior to placing of concrete. Only concrete mix designs approved by the customer shall be use.
- In miscellaneous works involving small quantities of concrete and where permitted by the Designer, one bagger mixer may be used. Concrete shall be mixed for at least 1-1/2 minutes after all the materials are in the mixer drum.
- Production method and facilities for ready mixed concrete shall conform to ASTM C – 94.

4.2 Formwork

Concrete shall be placed in forms, unless otherwise approved by the QC Team. Formworks shall be constructed in such a manner as to enable the forms to be removed without hammering or

prying against the concrete. Chamfers of 50mm are required to exposed exterior corner. Forms shall be of wood, plywood, steel or other approved materials, free from roughness and surface irregularities, substantially watertight and adequately braced and tied to prevent sag or movement when concrete is placed.

4.3 Placing of Concrete and Sampling

Inspection and approval by the QC Team will be required for the depth and character of the foundation, the adequacy of the forms and false work, and the placement of steel reinforcement and embedded items before concrete placement. Before depositing concrete all debris foreign matter, dirt and water shall be removed, surface of any concrete previously placed shall be cleaned and then brush with cement paste.

The method and manner of placing concrete shall be such as to avoid the possibility of segregation of concrete materials or displacement of the reinforcement. Concrete shall be deposited in a continuous operation as far as it is practicable to do so with concrete poured evenly on all sides in a manner that will avoid initial setting in any part of the work before fresh concrete is placed on it. Mechanical vibrator or other suitable tools shall be used to the extent necessary to obtain proper consolidation and finish of concrete; but care shall be taken to avoid segregation of aggregates by excessive vibration.

Sampling of concrete shall be in accordance with ACI code 211. Number of samples to be taken shall be based on project specification.

4.4 Protection and Curing

Newly placed concrete shall be cured and protected from dangerous elements and fast flowing water and from defacement. As soon as the concrete has sufficiently hardened, it shall be covered with damp, close woven burlap or similar materials, or with clean sand, which shall be kept thoroughly saturated with water over a period. When wood forms are used they shall be kept wet for same period to prevent openings at a joint and drying out of concrete.

4.5 Removal of Forms and Shores

Forms shall not be disturbed until the concrete has sufficiently hardened. Shoring shall not be removed until the supported concrete has acquired adequate strength to support its own weight and the load upon it. Concrete subject to additional loads during construction, shall be adequately shored to sustain all resulting stresses. Removal of forms shall be in accordance with ACI – 347 and shall be accomplished in such a manner and sequence as will prevent injury, forms may be required to remain in place for longer periods.

4.6 Concrete Finish

As soon as the forms have been stripped, form ties, if employed, shall be removed, and recesses left by the ties shall be filled with cement mortar. Any defect, such as honeycombs, in the surface of the concrete shall be chipped out and repaired in a suitable manner. Concrete surfaces that are not exposed in the completed work will require no special finish other than what is necessary to leave them smooth and impervious.

Concrete pouring and finishing shall be scheduled so that all finishing work done by capable and experienced men before the concrete has attained its final set.

4.7 Definitions & Terminologies

MES	Method Statement
WIR	Work Inspection Request
ITP	Inspection Test Plan
PM	Project Manager
CM	Construction Manager
QC	Quality Control
PPE	Personal Protection Equipment
HSE	Health Safety & Environment Plan.
TBT	Tool Box Talk
JSA	Job Safety Analysis
RAM	Risk Assessment Matrix

4.8 Initiation

4.8.1 TBT (Tool Box Talk):

a) Life Rule – Personal Protection Equipment (PPE)
Did You Know? - If your Personal Protection Equipment isn't exactly things can go seriously wrong.

- For many Task PPE is the only things that between you and injuries or worse, whether we're talking respirator, safety glasses, gloves, helmet or safety shoes, it's not enough to just identify the correct PPE for the hazard at hand. The equipment should be a proper fit and regular maintained and inspected before use. For example, hard hats should have HSE Sticker noting the date of inspection.

b) Life Rule – Permit to Work
Did you Know? – Allowing high risk work without permit is risky business

- Before starting any activities in the specific areas, secure PTW to proceed in the activity.
- High risk activities should have Hot Works PTW and Cold Work for the routine activities.

c) Safety Moment – Slip and Trip
Did You Know? – By following a few simple rules, you can avoid slips, falls, and tripping.

- Be mindful of your surroundings. From people, equipment, on-going operations then note the potential risks.
- Report all irregular surfaces, slipper areas, weak surface grounds, wrong placed of equipment and other unsafe conditions.
- Never use mobile phone walking and working.

d) Safety Moment – Heavy Lifting
Did You Know? – Heavy Lifting requires brain as well as muscle.

- Plan the lift and follow a JSA (Job Safety Analysis) as everyone aware on the activity around the lift area.
- For critical lifts, inspect the equipment, belts and other lift tools. Know the load weight and equipment capacity to load.
- Never walk or stand around the lifting area.

e) Safety Moment – Extreme Heat
Did You Know? – Working in a hot environment increases the risk of Heat related illness.

- When a person works in a hot environment, the body must reduce excess heat. Work safely by keeping yourself hydrated, wearing appropriate clothing / PPE and take the shelter to recover from heat expose.

4.8.2 Safety Precautions

a.) Ensure safety of personnel and the environment as per Project standards.

b.) All personnel involved will use necessary PPE as required such as safety helmet, glasses, coverall, shoes, gloves, ear plugs etc.

c.) The workers will made aware of the safety requirements related to this activity through daily toolbox and safety talk. The Contractor will instruct subcontractors to inspect all equipment prior to use.

d.) All required work permits will be acquired and kept available at respective work Site hanged clearly in plastic folders.

e.) Spark arrestor and warning lights will be installed on all mobile equipment.

f.) The subcontractor will inspect all equipment prior to use.

g.) Every effort shall be made to assure that employees can accomplish their assigned scope of work safely.

h.) The ability to determine the storage areas, access, potential and existing hazard on the will identified the facilities for welfare, first aid and sanitation, the work permit procedures and requirements and the basic precautions on dealing with fire hazard and damage to property at the planning stage.

i.) Ensure the proper sequence of construction operations, outline potential hazards at each stage and indicate the precautionary measures to be adopted on site by the area foreman / site supervisor. Carry out checking over work methods and precautions against supervision prior the commencement of work.

j.) Carry out provision of the warning sign and isolation requirements on site.

k.) Contractors and his subcontractors and sub-subcontractors shall fully comply with approved Project HSE Plan.

l.) The safety precautions are also listed in above Job Safety Analysis (JSA) and Risk Assessment Matrix (RAM).

4.9 Risk Assessment Matrix

What are the hazards?		Who might be harmed?	Control Measure	Risk Evaluation			Risk Rating
					Likelihood (1 – 3)		
Inexperienced and untrained personnel	Carrying out tasks without care due to insufficient	Worker	Read and sign code of practice for this lab. Receive training in safe use of equipment and methods.	2	2	4	Medium
Heavy concrete specimens and equipment	Lifting very heavy weights causing injury and spillage as	Worker	Store appropriately in storage or wheelie bins. Anything above 25kg should only be moved by one person using a wheelie bin or wheel barrow.	2	1	2	Medium
Overhead crane	Head injuries	Worker	Wear safety helmet, if overhead crane is active.	2	3	6	Low
Tripping	Injury from falling	Worker	Good housekeeping, do not leave anything lying around. Clean up spills immediately. Use wet floor	2	1	2	Low
Chemicals	Harmful to human health by inhalation	Worker	Minimise quantities stored in the lab and store in flammable solvent cabinet. Always wear gloves and lab	2	2	4	Medium

Wet concrete – Ready mixed, hand or machine mixed. Severe cement burns, skin irritation, cracked skin and contact/irritant dermatitis arising from skin contact	Harmful to human health by inhalation and skin contact.	Worker	Awareness and compliance of the contents and control measures of the material COSHH assessment for this product. Washing, changing and first aid facilities are available on site. Concrete soiled clothing must be changed	2	2	4	Medium
Dry Concrete – Bagged and hand or machine mixed. Respiratory problems to the mouth, throat and lungs, silicosis and fibrosis arising from the inhaling of airborne concrete dusts.	Harmful to human health by inhalation and skin contact.	Worker	Awareness and compliance of the contents and control measures of the material COSHH assessment for this product. Bags are issued, transported and stored carefully to avoid split bags. Eye, face, gloves and full clothing are worn. Mixing must be on a dry board and opened carefully avoiding spillage and airborne dust. Avoid mixing during exposed windy conditions and try to stand upwind with the face turned away	2	2	4	Medium

Working at Height	Noise	Hand tools	Vibration
Falling objects	Extreme Heat / cold	Confined spaces	Repetitive hand/ arm movement
Slippery/ uneven/ worn floors	Radiation	Poor housekeeping / cleaning	Machine operation
Obstructions/ projections	Lighting	Vehicle movement	Electro Magnet
Manual handling	Compressed air	Fire / explosion	Pressurised systems
Mechanical Lifting	Substances / materials	Electricity	**Other (*specify on assessment)***

2. RISK MATRIX		**Potential consequence of harm**		
		1 – Minor Injury (e.g. hazard can cause illness, injury or equipment damage but the results would not be expected to be serious)	**2 – Significant Injury** (e.g. hazard can result in serious injury and/or illness, over 3 day absence)	**3 – Major Injury** (e.g. hazard capable of causing death or serious and life threatening injuries)
Likelihood of harm	**1 – Unlikely** (injury rare, though possible) **2 – Possible** (injury could occur occasionally) **3 – Probable** (injury likely to occur, can be expected)		**4 – Medium**	

3. RISK EVALUATION

This is calculated by multiplying the likelihood against the consequence e.g. taking a likelihood of 1, which is classified as Unlikely and multiplying this against a Potential Consequence of 2, which is classified as Significant Injury, would give you and overall Risk Rating of 2, which would result in an overall evaluation as a low risk.

1 to 2 = **Low risk**

Low risks are largely acceptable, monitor periodically to determine situation changes which may affect the risk, or after significant changes

3 to 4 = **Medium risk**

Medium risks should only be tolerated for the short-term and then only whilst further control measures to mitigate the risk are being planned and introduced, within a defined time period.

6 = High risk

High risks activities should cease immediately until further control measures to mitigate the risk are introduced. The continued effectiveness of control measures must be monitored periodically.

9 = Extreme Risk

Work should not be started or continued until the risk has been mitigated. Immediate action is required to reduce exposure. A detailed mitigation plan must be developed, implemented and monitored by senior management to reduce the risk before work is allowed to commence

4.10 Authorization and Distribution

Managerial Contact

Name	Contact Details

Engineers and Supervisor Contact

Name	Contact Details

Technician and labor involve.

Task	Trade	Number

4.11 Document Control

Content	Reviewed			Document Control Record
	Construction Supervision	Quality	HSE	
Scope of works				
Purpose				
Responsibilities				
a.) Managerial staff				
b.) Engineer & Supervision staff				
c.) Construction labour				
Installation Procedure				
Safety risks issues				
Safety precautions				
Inspection parameters				

4.12 Communications

To ensure adequacy of methods, correct level of managerial and supervision support, appropriate safety measures for the risks identified, correct trades are used, and the appropriate equipment and tooling is used. Any concerns are to noted and resolved prior to commencement of the works.

5. Guidance Notes and Standard Forms

- Work Permit
- Scaffolding Tag
- Work Inspection Request
- Checklist – Cable tray/Cable Ladder

6. References

- Shop Drawings
- ITP
- Material Inspection Report
- Project Specification
- Quality Plan
- HSE Plan